

The Hand That Gives

Pre-PIE *de-* and the Polyfunctional Architecture: a Restoration of
Marstrander's 1919 Comparison on the Heteroclitic Foundation

Paper 1 of the der- project

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Abstract

A comparative cluster first proposed by Marstrander in 1919 — Hittite *dār-na-áš* (a measure-noun derived from the Akkadogram UP.NI "fist"), Old Irish *dorn*, Welsh *dwrn*, Gaulish *durno-*, Old Irish *derna* "palm," Greek δῶρον "palm of hand; gift," and Greek δάρις (Hesychius) — was incorporated into Pokorny's *IEW* (1959: 203) and remained in the standard handbook until late-twentieth-century skepticism dismantled it. Schrijver (1995) labeled the δωρ-/δαρ- relation "obscure"; Beekes (2010) excluded the Celtic forms from δῶρον on grounds of u-vocalism and separated the gift and palm-breadth senses into unrelated entries; Matasović (2009) classified Celtic *durno-* as probably non-IE in origin. The present paper restores Marstrander's analysis on the formal foundation that was unavailable to him: Clayton's (2023) reconstruction of the **-wṛ̥-/w(e/o)n-* heteroclitite class, with which Blažek (1999) had independently begun to formalize the same cognate set, and the parallel decomposition of *de-* as a free pre-PIE morpheme that Papers 2–4 of this project develop. Under this reading, the cluster reflects two derivational outputs on a single pre-PIE element *de-* "open hand": a locative-extended body-part noun *de-r-* (yielding δάρις, *derna*, Continental Celtic *DERNV*[]) and a heteroclitic gift-noun *deh₃-wṛ̥-/w(e/o)n-* (yielding δῶρον, Latin *dōnum*, Celtic *durno-*, Latvian *dūre*, Anatolian *tarawar*). The Celtic *u* that frustrated Beekes is the regular reflex of the Proto-Nuclear-IE metathesis **wṛ̥ > *ru > *ur*. The framework also recovers Albanian *dorë* "hand" as a member of the cluster, supported by a 127-year published lineage (Meyer 1883 → La Piana 1937/1942 → Belardi 1948 → Pokorny 1959 → Schindler 1967 → Frisk → Beekes → DGE) that Demiraj (1990, 1997) reversed on grounds that do not survive examination. Probability: 40–50%.

1. Introduction

The comparative cluster under discussion has been recognized as a unit since Marstrander assembled it in 1919, within four years of Hittite's decipherment

by Hrozný. On page 128 of *Caractère indo-européen de la langue hittite*, in the course of arguing that Hittite *a* in *dār-na-áš* descends from IE **o*, Marstrand identified the Hittite measure-noun (alternating with the Akkadogram UP.NI "fist, hand, handful") with Old Irish *dorn*, Welsh *dwrn*, and Gaulish *durno-*, and in footnote 1 he extended the comparison to Old Irish *derna* "palm" (in apophonic alternation), to Greek δῶρον (with the Hesychian gloss παλαιστή "palm-width"), and to Greek δάρις (with the Hesychian gloss σπιθαμήν "span"). The cluster is presented with explicit ablaut analysis, with the Hesychian apparatus, and as evidence for a sound law: Marstrand offered it not as an etymological aside but as structural support for his phonological argument about Hittite vocalism.

Pokorny incorporated Marstrand's cluster into *IEW* 203 in 1959, classifying the forms under **der-* "Spanne der Hand," and the comparison remained the standard handbook treatment for the next half-century. Three late-twentieth-century departures from this lineage motivate the present paper.

Schrijver (1995: 58), listing the forms as "possible cognates," described the δῶρ-/δάρ- relation as "obscure" — a characterization that accurately reflected the absence of a satisfactory formal account within standard PIE phonology, but that has since been read by some as a substantive rejection of the comparison itself.

Beekes (2010), in the *Etymological Dictionary of Greek*, noted that Celtic "points to PIE **u*, which excludes any relationship with δῶρον," and on this basis separated the "gift" and "palm-breadth" senses of δῶρον into unrelated entries. The phonological problem was real: the Celtic *u* is attested in Continental Gaulish from the first century BCE (*Durnomagus*, *Durnacos*, *Dagodurnus*) and predates any Welsh-internal raising rule (Ball & Müller 2009), so it must be original at the Proto-Celtic level. What Beekes lacked was a formal mechanism to derive the *u* from the same etymon that gives δῶρον.

Matasović (2009, s.v. *durno-*) drew the appropriate methodological conclusion: without a viable IE etymology for the Celtic *u*, *durno-* should be classified as probably non-IE in origin. The substrate hypothesis was a disciplined response to the absence of a satisfactory formal account, not a dismissal of the comparison.

The present paper proposes that a viable IE etymology is now available. Clayton (2023), in a UCLA dissertation, systematically reconstructs a class of PIE neuter heteroclites with strong stems in *-wṛ-* and weak stems in *-w(e/o)n-*, and demonstrates that a regular metathesis **wṛ* > **ru* operated at Proto-Nuclear-IE depth — after the Anatolian split, before branch separation. Critically, Clayton independently identifies **deh*₃- as a root that forms

heteroclitics of this class. Blažek (1999), publishing in *Tocharian and Indo-European Studies*, had independently arrived at the same heteroclitic structure for this cognate set, formalizing Marstrander's comparison as **dwor-eH₂-wr̥-/w(e)n-* "handfulness" and adding the Anatolian witness *tarawar* "measure of capacity" (with the corresponding verb *tarāwi(ya)-* "to hand over, deliver"). The framework of the present paper unites Blažek's and Clayton's reconstructions on a single decomposition: the cluster reflects two derivational outputs on a pre-PIE element *de-* "open hand," developed across Papers 2–4 of this project as a free morpheme productive across multiple grammaticalization pathways.

Under this reading, Marstrander's **o*-grade and **e*-grade ablaut becomes the surface manifestation of two distinct derivational operations on a common base. The locative extension **de-r-* yields the body-part noun attested in Hesychian δάρις, Old Irish *derna*, and Continental Celtic *DERNV*/. The heteroclitic extension **deh₃-wr̥-/w(e/o)n-* yields the gift/measure-noun family attested in δῶρον, Latin *dōnum*, Celtic *durno-*, Latvian *dūre*, and Anatolian *tarawar*. The Celtic *u* is the regular outcome of the **wr̥ > *ru > *ur* metathesis-and-revocalization pathway documented by Clayton; the δωρ-/δαρ-alternation that Schrijver labeled "obscure" is the strong/weak stem contrast of a single heteroclitic paradigm.

The paper restores Marstrander's lexical comparison on this formal foundation, recovers Albanian *dorë* "hand" as a member of the cluster (with the 127-year published lineage that supports it), and integrates the Anatolian witness **tarawar*/**tarāwi(ya)-* that Marstrander already had in *dār-na-áš*. The framework's specific contributions beyond Marstrander are: the heteroclitic formalization of the gift-noun branch (jointly Blažek/Clayton); the bimorphemic decomposition of *de-* + *-r-* and *de-* + *-h₃-* (Papers 2–3); the locative status of *-r-* (Lipp 2009, Bauhaus 2019, Kahl 2024, all independent); the Albanian *dorë* extension (§8 below); and the integration with Norbruis's (2021) chronology of the give/take semantics (§9).

The paper is organized as follows. §2 sets out the polyfunctional architecture of the broader four-paper project. §3 presents Marstrander 1919's comparison in detail, with attention to its ablaut analysis and Hesychian apparatus. §§4–7 develop the framework's analysis of the body-part noun, the heteroclitic gift-noun, the Celtic metathesis pathway, and the δωρ-/δαρ-alternation. §8 develops the Albanian *dorë* extension. §9 addresses the Anatolian evidence and the chronological architecture. §10 engages the late-twentieth-century departures from the published lineage. §11 sets out scope and limitations. §§12–13 close with the remaining papers and the probability assessment.

2. The Polyfunctional Architecture

The framework rests on the reconstruction of a pre-PIE element *de-* "hand" that was productive across multiple derivational pathways simultaneously. Body-part terms are cross-linguistically among the most fertile sources of grammaticalization and morphological productivity: the hand reaches, points, conveys, spans, and gives, and these functions leave distinguishable lexical and grammatical residues. Heine and Kuteva (2002) document the resulting range of pathways — HAND > LOCATIVE, HAND > AGENT MARKER, HAND > GIVE, HAND > DIRECTIONAL — across typologically diverse languages. The present proposal is that pre-PIE instantiated several of these pathways in parallel, and that the comparative evidence in IE reflects the different outputs.

The full architecture of the four-paper project is shown in Table 1. The present paper develops the first two rows; Papers 2–4 develop the remaining branches. The reconstruction does not require any new PIE roots: the bare *de-* and the locative *-r-* are independently attested morphemes, and the framework is a proposal about their prior productive relationship, not an introduction of new material.

Root + suffix	Surface form	Meaning	Paper
*de- + *-r-	*de-r-	body-part noun (δάρις, <i>derma</i> , <i>DERNV</i> [])	1
*de- + *-u̯-/-w(e/o)n-	*deh ₃ -w̥-	gift/measure-noun (δῶρον, <i>dōnum</i> , <i>durno-</i> , <i>tarawar</i>)	1
*de- + *-r- + *-o-	*(t)ero-	bilateral-contrastive suffix	2
*de- + *-u̯- + *-o/-i-	*d̥u̯-o-/d̥u̯-i-	numeral 'two'	2
*de- + *-h ₃ -	*deh ₃ -	give-verb (post-Anatolian)	3
*de- + postposed *-a	*de + a → *dā-	'take' (Anatolian)	4

Table 1. Derivational pathways on pre-PIE **de-* across the four-paper project.

Two features of the skeleton bear on the present paper's central claims. First, the locative *-r-* that appears in both the body-part noun (row 1) and the contrastive suffix (row 3) has independent support from three published treatments: Lipp (2009), reconstructing the PIE heteroclitic *r/n*-stem paradigm, treats *-r* and *-én* as parallel locative endings; Bauhaus (2019) argues that *-r* is a true PIE locative case ending rather than an adverbial particle; and Kahl (2024), working on degree morphology, independently derives the secondary comparative of Indo-Iranian and Greek from "an archaic locative morpheme **-er*." The convergence of three independent treatments from three scholarly

traditions is the strongest available warrant for treating *-r-* as a productive locative morpheme in early PIE.

Second, the bare *de-* directional particle is not a theoretical abstraction: it survives as the Greek allative clitic *-δε* (productive in Mycenaean by 1400–1200 BCE; Lejeune 1961, van Hes 2021), as Old Church Slavonic *do*, as Proto-Germanic **tō* (English *to*, German *zu*), and in Celtiberian *DV* (Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45). Paper 3 develops these as the grammaticalized descendants of a pre-PIE hand-element via the documented HAND > DIRECTIONAL pathway. Paper 4 develops the further claim that Hittite *dā-* "to take" preserves a fossilized univerbation *de + a* in which the bare hand-element combined with the postposed directional particle *-a* and contracted by hiatus.

3. Marstrander 1919: The Original Comparison

Marstrander's comparison occupies page 128 of *Caractère indo-européen de la langue hittite*, in the section establishing that Hittite *a* descends from IE **o* in certain positions. The relevant passage reads:

Un exemple certain d'i.-e. **o* est présenté par l'hitt. *dār-na-áš*, qui selon M. Hrozný (p. 41, note 2) Ks II 4 n° 4 alterne avec accad. UP.NI «poing, main, poignée». Je l'identifie avec le thème en **o* irl. *dorn*, gallois *dwrn*, gaulois *durno-* «poing, poignée, largeur d'un poing (mesure)».¹

Translation: "A certain example of IE **o* is presented by Hittite *dār-na-áš*, which according to Hrozný alternates with Akkadian UP.NI 'fist, hand, handful.' I identify it with the **o*-stem Irish *dorn*, Welsh *dwrn*, Gaulish *durno-* 'fist, handful, breadth of a fist (measure).'"

Footnote 1 extends the cluster:

En rapport apophonique avec ce thème est celui en *n*: irl. *derna* «la palme». D'autres degrés vocaliques se trouvent dans δῶρον· παλαιστή, δάρις· σπθαμήν (Hésyche).

Translation: "In apophonic relation with this theme is the one in *n*: Irish *derna* 'the palm.' Other vocalic grades are found in δῶρον· παλαιστή, δάρις· σπθαμήν (Hesychius)."

Three features of Marstrander's analysis bear on the framework's reconstruction.

First, the comparison is structural rather than lexical-isolated. Marstrander offers it as evidence for a sound law (Hittite *a* < IE **o*), with the cognate set doing structural work in his phonological argument. This is published Indo-Europeanist analysis of a kind that would not have been advanced as

speculative aside.

Second, the ablaut analysis is explicit. Marstrander distributes the cluster across distinct grades: the *o-grade theme contains the Hittite measure-noun and the Celtic forms; an *n-stem theme contains Old Irish *derna* "palm"; further unspecified vocalic grades contain Greek δῶρον and δάρις. The framework's *e-grade *de-r- (body-part noun) and *o-grade gift-noun analysis is a development of Marstrander's original distribution, with the locative -r- (Lipp/Bauhaus/Kahl) and the heteroclitic -wṛ-/w(e/o)n- (Blažek/Clayton) supplying the morphological mechanisms that were unavailable in 1919.

Third, the Hesychian apparatus is in place from the start. Marstrander cites Hesychius for both Greek forms, with measurement-related glosses: δῶρον is glossed παλαιστή (the Greek measure for palm-width, four finger-breadths), and δάρις is glossed σπιθαμήν (the standard Greek "span"). The Vitruvian connection between δῶρον "gift" and δῶρον "palm" (De architectura 2.3.3, on which §7 below) is corroborated by Hesychian lexicography that Marstrander already had in evidence.

Marstrander's *dār-na-áš* corresponds to what the standard Hittite etymological dictionaries now treat as the measure-noun *tarna-* (Tischler HEG III, Lieferung 9, p. 195), distinct from the verb *tarna-* "to release, let go." The framework's connection to the verbal "hand-over" semantics runs through the related Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" and the corresponding noun *tarawar* "measure of capacity" (Tischler 1991: 155–156; Melchert 1993: 211; Blažek 1999), discussed in §9 below.

4. The Body-Part Noun *de-r-*

The framework's *e-grade analysis treats Marstrander's *n-stem *derna* as the *e-grade output of a locative-extended hand-noun *de-r-*. The locative status of -r- rests on three independent published treatments.

Lipp (2009), reconstructing the PIE heteroclitic r/n-stem paradigm, argues that the locative singular forms of acrostatic heteroclitite neuters are built from root nouns combined with an adverbial particle: *-r and *-én as parallel locative endings meaning "at/in the place of (X)." On Lipp's analysis, the spatial semantics is primary and the body-part association is downstream.

Bauhaus (2019) advances a stronger version: *-r is not an adverbial particle but a true PIE locative case ending — "a real case ending, not an Adverbialendung" — active in pronominal and nominal formations across the family: *k^wor (Gothic *þor* "where"), *tor (Gothic *þar* "there"), nominal locatives such as *ǵ^hǵm-ér "on the earth."

Kahl (2024), studying degree morphology in a Harvard dissertation under Jeremy Rau, reaches the same conclusion from a third direction: "the origin of the 'secondary comparative' of Indo-Iranian and Greek is traced to an archaic locative morpheme *-er."

The three treatments are methodologically independent; none is building on the others. The compositional reading of *de-r-* — "at the place of the hand-direction" — is the natural semantics given the attested body-part function of the forms.

The cluster's surface witnesses to the *e-grade body-part noun:

Hesychian δάρις / δάρειρ, glossed σπιθαμή ("span of all fingers"). The form is an i-stem derivative of *de-r-*, morphologically transparent: the open palm as a locative body-part anchor, "the span at the place of the hand-direction." Marstrander 1919 already cited this gloss in his footnote 1.

Old Irish *derna* (Gaelic *deàrna*) "palm of the hand." Continues the unmarked feminine singulative *de-r-nā-*. Marstrander 1919 identifies this as the *n-stem in apophonic alternation with the *o-grade *durno-*.

Continental Celtic *DERNVI* at the Bronze of Novallas (Stifter 2022) — a Celtiberian hospitium bronze that preserves the body-part noun alongside the directional particle *DV* (Paper 3's Celtiberian reflex of *de/do*). Two derivational branches of the bare hand-element appear in a single inscription, a convergence that the existing literature does not note since it is visible only through the framework's lens.

Bauhaus's most structurally significant contribution for the broader framework is his reconstruction of a paradigmatic opposition between locative *-r and directive *-o attached to the same nominal roots: *h₁upér "on top" vs. *h₁upó "to the top," *k(e)nter vs. *knto, *ndher vs. *ndho. This paradigmatic pair is the structural template the framework reconstructs on *de-: locative *de-r- (the body-part noun) versus directive *de + a (Paper 4's analysis of Hittite *dā-*). Bauhaus's observation that the locative *-r "ceased to be productive in all attested languages" while the directive member "could keep its productivity at least until Anatolian" directly supports the chronological asymmetry the framework reconstructs.

5. The Heteroclitc Gift-Noun

The δωρ-/δαρ- alternation that Schrijver labeled "obscure" and the Celtic *u* that Beekes treated as exclusionary can be addressed together, through Clayton's (2023) reconstruction of the PIE *-w̥r-/w(e/o)n- heteroclitc class.

Clayton's dissertation systematically reconstructs a class of neuter heteroclites with alternating strong stems in $-w\check{r}_-$ and weak stems in $-w(e/o)n-$, and demonstrates that a regular metathesis $*w\check{r}_- > *ru$ operated at Proto-Nuclear-IE depth — after the Anatolian split, before branch separation. Critically, Clayton independently identifies $*deh_3-$ as a root that forms heteroclites of this class, with Sanskrit *dārú-* "liberal" (from $*déh_3-w\check{r}_-$) and Prasun *pyörü* "gift" (from $*pro-deh_3-w\check{r}_-$) among his examples. His argument is not concerned with Celtic or with the etymology of $\delta\omega\rho\upsilon\nu$; the application to those forms is the contribution of the present paper.

Blažek (1999), in *Tocharian and Indo-European Studies* 8, had independently arrived at a heteroclitic structure for the same cognate set, formalizing the cluster as $*dwor-eH_2-w\check{r}_-/-w(e)n-$ "handfulness" and including Anatolian *tarawar* (the Hittite/Luwian measure-noun discussed in §9 below). Blažek's reconstruction differs from the framework's in one notational respect — he writes $*dw-$ with consonantal *w*, while the framework treats this as $*de-$ + $*-u-$ under Paper 2's decomposition of $*d\check{u}o-$ "two" — but the underlying morphological architecture is the same: a heteroclitic $*-w\check{r}_-/-w(e)n-$ formation on a $*d(e)-r/- (u)-$ base, ablauting across the cluster.

With this framework in place, the derivation of $\delta\omega\rho\upsilon\nu$ from the heteroclitic strong stem is as follows: PNIE $*deh_3-w\check{r}_-$ undergoes the regular metathesis to give $*deh_3-ru-$; the laryngeal $*h_3$ colors and lengthens the preceding vowel, giving $*dō-ru-$; thematization yields $*dōr-o-n$; Greek surfaces as $\delta\omega\rho\upsilon\nu$. The standard exemplar of the $*w\check{r}_- > *ru$ metathesis is $*k^wetw\check{r}_-$ "four" $\rightarrow$ $*k^w(e)tru-$ in Avestan *caθru-*, Gaulish *petru-*, Latin *quadru-* (Clayton 2023: §2.2). The mechanism is well-attested and uncontroversial; its application to $*deh_3-$ follows from Clayton's independent identification of this root as a member of the relevant class.

The weak stem accounts for the remaining Greek and Latin forms. The heteroclitic weak stem $*deh_3-w(o)n-$ (zero-grade $*dh_3-w(o)n-$) yields Greek $\delta\acute{\alpha}\nu\omicron\varsigma$ "gift, loan" with regular zero-grade vocalization. Latin *dōnum* is the same weak stem, thematized, with the long $\bar{o}$ from laryngeal coloring in the full grade.

Form	Language	Stem	Analysis
$\delta\omega\rho\upsilon\nu$	Greek	strong, metathesized	$*deh_3-w\check{r}_- > *dō-ru-$
$\delta\acute{\alpha}\nu\omicron\varsigma$	Greek	weak, zero-grade	$*dh_3-w(o)n-$
<i>dōnum</i>	Latin	weak, full grade, thematized	
<i>dārú-</i>	Sanskrit	strong, metathesized	Clayton (2023)

<i>durno-</i>	Celtic	strong, metathesized + revocalized	§6 below
<i>dūre</i>	Latvian	strong, Baltic development	Schrijver (1995)
<i>tarawar</i>	Hittite/Luwian	heteroclitic, full reconstruction	Blažek (1999); §9 below

Table 2. The heteroclitic cognate set under the framework's analysis, restoring Marstrander 1919's comparison.

6. Celtic *durno-* and the Metathesis Pathway

The Celtic derivation requires separate treatment because of the original *u* (the formal point that Beekes 2010 found exclusionary) and because the compound evidence bears on the base meaning.

The **u**-vocalism is attested in Continental Gaulish from the first century BCE — *Durnomagus*, *Durnacos*, *Dagodurnus* — predating any Brythonic sound change. The chain from PNIE **deh₃-wr̥-* to Celtic **durno-* runs as follows: the PNIE metathesis **wr̥ > *ru* gives **deh₃-ru-*; in non-final syllables adjacent to a labial **w*, the syllabic resonant **r̥* is **u**-colored, yielding **dur-* (the Lubotsky mechanism, documented for Greek and Indo-Iranian in Lubotsky 1994, 1997); a singulative suffix **-no-* then gives **durno-*, after loss of the laryngeal in pre-Celtic. The closest phonological analog in the wider literature is OKhotanese *durna-*, which Bailey (1961) traces from PIIr. **druna-* by the same metathesis and revocalization; Clayton (2023) cites this parallel for his Sanskrit data, and it is equally the best available analog for the Celtic development.

The compound evidence bears on the base meaning. Proto-Celtic **are-durnon* (Old Irish *aurdorn*, Welsh *arddwrn*, Breton *arzorn*) means "wrist"; **ambi-durnos* (Old Irish *imdorn*) means "sword hilt." Neither compound is semantically natural with "fist" as the base; both are natural with "hand" construed as an anatomical unit. Breton reinforces this reading synchronically: *dorn* forms the body-part dual *daouarn* (= *daou* "two" + *dorn*), a category Breton reserves for anatomically paired body parts (Ball & Müller 2009: 445–446). The derivation *dorn* → *dornad* "handful" (with the suffix marking container contents) likewise presupposes an open-hand reading of the base.

The semantic evidence from Celtic compounds and derivatives is consistent with a body-part noun meaning "hand" rather than a posture term meaning "fist," which aligns with Marstrander's distribution: the **o*-grade **durno-* covered the hand as an anatomical unit, with "fist" as the contextually salient configuration in martial and combative contexts. Old Irish and Welsh narrowed to "fist" because the martial context dominated in the Insular Celtic record;

Breton and the Continental compounds preserve the broader "hand" reading. This is exactly the kind of polysemy that body-part nouns commonly show.

7. The δῶρον-/δάρις- Alternation and the Vitruvius Testimony

The alternation between δῶρον and δάρις — documented by Schrijver as formally obscure, recognized but unanalyzed by Marstrander — becomes transparent under the heteroclititic analysis: δῶρον is the thematized strong stem (**deh₃-wr̥-* > **dōru-*) and δάρις is an i-stem derivative of the bare body-part noun (**de-r-*). They are not the same surface formation and need not exhibit the same vocalism; the connection is family relationship through the shared pre-PIE base.

Beekes's decision to separate the "gift" and "palm-breadth" senses of δῶρον into unrelated entries addresses a real formal problem within the standard reconstruction. The framework offers an alternative under which both senses descend from the same hand-root, one through the gift-noun branch of the heteroclitite and one by family association with the body-part noun. That the two senses were understood as connected in antiquity is confirmed by Vitruvius, who states explicitly that the Greeks call the palm δῶρον because giving (*datio*) is called δῶρον in Greek, and a gift is always conveyed through the palm of the hand (De architectura 2.3.3).

Vitruvius's testimony does not settle the etymological question — an ancient author's folk etymology is not a formal derivation — but it is evidence that the semantic connection between the two senses was transparent to Greek speakers. Marstrander's Hesychian glosses point in the same direction: δῶρον · παλαιστή connects the gift-noun to the palm-width measure that Vitruvius describes; δάρις · σπιθαμήν connects the body-part i-stem to the standard Greek "span." The lexicographic and architectural traditions converge on a connection between the two senses that the framework explains as etymological family relationship rather than accidental polysemy.

8. Albanian *dorë*: The Framework's Principal Extension

Beyond Marstrander's cluster, the framework recovers Albanian *dorë* "hand" as a member of the comparative family. The recovery rests on a 127-year published lineage that Demiraj (1990, 1997) reversed on grounds that do not survive examination, and on a recent reconstruction in DGE that essentially identifies the framework's **de-h₃-r-* in different notation.

8.1 The published lineage (1883-2010)

The connection of Albanian *dorë* with Greek δῶρον was first proposed by Gustav Meyer in 1883 (*Albanesische Studien* I, p. 92). Marco La Piana developed the

phonological argument in two publications: a 1937 monograph *Intorno al riflesso della vocale "o" lunga dall'indoeuropeo e dal latino nell'albanese* (Palermo) and a 1942 short note in *Indogermanische Forschungen* 58, p. 98. Walter Belardi's 1948 article in *Rivista degli Studi Orientali* 23, pp. 69–81 ("Ittito *ki-eš-šar*, greco χείρ") argued on independent grounds that the standard reconstruction **ĝ^hesr-* "hand" cannot yield Albanian *dorë* — the *-sr-* cluster would give Albanian **dostrë* by regular sound change, with the Albanian parallels *shtërpinj* and Liburnian *Εἴστρις* supplying the comparative basis. Pokorny propagated Belardi's argument in *IEW* 447, eleven years later, listing the explicit phonological output (**dostrë*) and concluding that Albanian *dorë* requires a non-s form. On p. 203, Pokorny placed Albanian *dorë* under **der-* "Spanne der Hand" alongside Greek δάρις and δῶρον, citing La Piana 1942 as the supporting reference. Schindler (1967, "Tocharische Miscellen," *IF* 72: 244) supported Pokorny's doubts about **ĝ^hesr-*.

The connection survives in three modern standard Greek etymological dictionaries. Frisk's *Griechisches Etymologisches Wörterbuch* (1960–72) preserves La Piana's **dōrom* > Albanian *dorë* derivation in the entry for δῶρον "Handbreite." Beekes's *Etymological Dictionary of Greek* (2010: 363) cites La Piana for the same connection. The *Diccionario Griego-Español* (DGE, online) reconstructs the body-part **deh₃r-* with explicit laryngeal **h₃* — essentially identifying the framework's **de-h₃-r-* in all but decomposition.

The lineage spans 127 years and seven major Indo-Europeanists or standard reference works: Meyer 1883 → La Piana 1937 → La Piana 1942 → Belardi 1948 → Pokorny 1959 → Schindler 1967 → Frisk 1960–72 → Beekes 2010 → DGE.

8.2 The Demiraj reversal and its grounds

Demiraj's *Albanische Etymologien* (1997: 140) reverses this lineage. The relevant paragraph reads:

"Der bisher ausschließlich angenommene Ansatz einer dehnstufigen Bildung nur für den alb. Vertreter wird von Pokorny (447: Grf. **ĝ^hes(o)r-* 'Hand'; s. auch Schindler *IF* 72 [1967] 244) in Zweifel gezogen, der offenbar nur aus diesem Grund den Zusammenhang Meyers (AS I 92) mit gr. δῶρον 'Handfläche' vorzieht (zustimmend La Piana *Intorno* 29ff.; Pok. 203: **der-* 'Handfläche'). Diese Deutung ist mit Tagliavini (AD 101) auszuschließen."

Demiraj acknowledges the Pokorny-Meyer-La Piana-Schindler lineage but excludes it on the authority of one source: Tagliavini *L'albanese di Dalmazia*, p. 101.

Direct examination of Tagliavini's *L'albanese di Dalmazia* (Olschki, Florence 1936) page 101 shows that page contains entries for *dëkum* "morto," *dëlme*

"pecora," *dër* preposition "fino, mentre," *dër* (-a) "porta," *dërdh* verb "versare, spandere," and *dërdhum* "diarrea, erpete." The page does not contain any discussion of Albanian *dorë* "hand," any mention of Greek δῶρον, any mention of **der-* "Handfläche," or any rejection of any alternative etymology for *dorë*.

Tagliavini's actual *dor* entry, on AD 106, contains six lines citing Meyer, Brugmann, Walde-Pokorny, Pedersen, and Thumb, and never engages the **der-/δῶρον* alternative.

The Demiraj rejection thus rests on a citation that does not deliver the rejection. Pokorny's own pages (203 favoring **der-*; 447 ruling out **ǵhesr-* on phonological grounds) contradict the **ǵhesr-* alternative that Demiraj endorses. The Pokorny analysis Demiraj defends is one that two named scholars (Belardi 1948 and Pokorny 1959) explicitly held to be phonologically impossible for Albanian.

8.3 The framework's specific claim

Albanian *dorë* descends from a lengthened-grade body-part formation **dē-r-* (from the framework's **de-r-*), with the Albanian *o* < ***ē* by the regular reflex documented in La Piana 1937 and Pokorny IEW 203. The framework adds two refinements to the standard analysis:

First, the lengthened-grade **dē-r-* is phonologically cleaner than the **dōrom* that the standard handbooks reconstruct (where the **o**-grade and the laryngeal create complications), and it brings Albanian *dorë* into direct alignment with the framework's **de-r-* base for the body-part noun.

Second, DGE's published reconstruction **deh₃r-* (with explicit laryngeal **h₃*) is essentially the framework's **de-h₃-r-* in different notation. The framework's specific contribution is to identify the **-h₃-* in DGE's body-part reconstruction with the **-h₃-* in PIE **deh₃-* "to give" — the morphological connection that DGE does not draw but that Paper 3 develops.

The Imberciadori (2025) supporting parallel, the Vitruvian semantic confirmation, and the Hesychian glosses in Marstrand 1919 are corroborative; the lineage from Meyer 1883 through DGE 2010 is the principal warrant. The framework recovers Albanian *dorë* as a witness to the cluster on grounds that have been continuously published for 127 years and that the Demiraj reversal does not adequately address.

9. Anatolian Evidence and Chronological Architecture

Marstrand's *dār-na-áš* corresponds to what Tischler's *Hethitisches Etymologisches Glossar* (1983, Lieferung 9) treats as the measure-noun *tarna-* — a unit equal to half an *upnu* (Akkadian for "clenched fist," cf. Carruba StBoT

2, 1966, 11 Anm. 5), distinct from the Hittite verb *tarna-* "to release, let go." The framework's connection to verbal "hand-over" semantics runs through a related Hittite/Luwian etymon that Tischler treats in a separate *Lieferung*.

Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" (Tischler 1991: 155–156; Melchert 1993: 211) and the corresponding noun *tarawar* "measure of capacity" (originally "handful, fistful") provide the Anatolian witness to the framework's heteroclitic gift-noun branch. Blažek (1999) reconstructs **T(e/o)reH₂-wr̥-* ~ *-wen-* for *tarawar*, explicitly invoking the same heteroclitic **-wr̥-/-(e/o)n-* structure that Clayton (2023) reconstructs more broadly. Blažek's full reconstruction:

**dwor-eH₂-wr̥-/-(e)n-* "handfulness" > Anatolian **tarawar/-wan-* "handful"
> "measure of capacity"; **dwōr-o-* > Greek δῶρον; **dur-o-unt(s)/-went-* >
Tocharian B *trau* "measure of capacity"; **dur-no-* > Celtic **durno-*; **dūr-iyā*
> Latvian *dūre*.

The framework's reading aligns with Blažek's substantive reconstruction. The notational difference — Blažek's **dwor-* versus the framework's **de-* + **-u-* + **-r-* via the Paper 2 decomposition of **d̥uo-* "two" — does not affect the surface forms or the heteroclitic structure. The framework's specific contribution is the morphological decomposition that connects this cluster to the broader architecture of Papers 2–4: the **-u-* of **d̥uo-* is the same **-u-* that figures in the heteroclitic **-wr̥-/-(e/o)n-*, and the bilateral semantics of the numeral and the contrastive suffix **(t)ero-* are inherited from the same pre-PIE element.

The Anatolian witness **tarawar*/tarāwi(ya)-* provides three substantive supports for the framework that were not available in the v33 draft.

First, it grounds the heteroclitic gift-noun branch in published Anatolian morphology (Melchert 1994: 14, 86; concerning the Anatolian suffix **-āwar* and its parallels, including Hittite *karawar* "horn" < **kreH₂-wr̥-*).

Second, it supplies the Anatolian "hand-over" verb that the framework's chronological architecture predicts. Norbruis (2021: 243–249) argues, on Anatolian-internal grounds, that the original meaning of **deh₃-* was "to take" rather than "to give," with the give-meaning a post-Anatolian innovation through the semantic chain "to take" > (with goal/beneficiary) "to bring, convey, deliver" > (with animate recipient) "to hand over, to give." The intermediate stage — physical delivery of concrete objects to animate recipients — is exactly the meaning of Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" (Tischler 1991). The Anatolian verb preserves Norbruis's reconstructed intermediate stage, in published form, on a heteroclitite of the framework's reconstructed root family.

Third, it converges with Marstrand's original cluster: Marstrand 1919 had the Hittite measure-noun *dār-na-áš* (= *tarna-*); Blažek 1999 added the Hittite/Luwian "handful" measure-noun *tarawar* with its corresponding "hand-over" verb *tarāwi(ya)-*. The Anatolian witness to the framework is now substantially fuller than Marstrand's single form.

The chronological architecture is coherent. Body-part noun first (pre-Anatolian, **de-r-*); directional particle second (pre-Anatolian, **de/do*); take-verb third (pre-Anatolian, Anatolian branch, **dā-*); give-verb and gift-noun fourth (post-Anatolian, PNIE, **deh₃-* and **deh₃-wr₆-*). The absence of the **deh₃-wr₆-* gift-noun from Anatolian as a verb of giving is expected: Anatolian preserves the root in *tarāwi(ya)-* "to hand over, deliver" at Norbruis's intermediate stage, with the abstract "gift" semantics post-Anatolian. Paper 4 develops the formal account of Hittite *dā-* "to take" as the Anatolian representative of the take-verb branch, deriving it from a pre-Anatolian directive univerbation **de + a* that resolves the take/give polarity structurally.

10. Previous Accounts: The Late-Twentieth-Century Departures

With the published lineage in place (§§1, 3, 8), the framework's engagement with the standard view is best framed as a return to the pre-Beekes consensus rather than as a refutation of established scholarship. Three departures from the lineage warrant brief comment.

Schrijver's (1995: 58) "obscure" assessment accurately described the absence of a satisfactory formal account within standard PIE phonology. The framework does not so much correct Schrijver as supply the formal tool that was missing: the Clayton (2023) heteroclite framework was not available at the time of Schrijver's writing, and Blažek's (1999) parallel reconstruction had not yet appeared when Schrijver composed his treatment. Schrijver's parenthetical observation, citing Beekes, that "in the Greek forms, δ- might reflect **du-* (cf. δωδεκα '12' < **duō-*)" anticipates the morphological-identity argument developed in Paper 2, where the numeral **duo-* is reconstructed as a derivational sibling of the contrastive suffix **(t)ero-* on the same root.

Beekes's (2010) separation of the two senses of δῶπον and his exclusion of Celtic on grounds of **u**-vocalism are both formally motivated within the evidence available to him. Beekes wrote without access to Clayton 2023 and without reference to Blažek 1999 (which sits in a Tocharian-studies venue). The framework offers an account under which the two senses are related and the Celtic *u* is the regular outcome of the **wr₆ > *ru* metathesis. The two accounts make different predictions about the morphological history and can be evaluated independently.

Matasović's (2009) substrate hypothesis followed from the absence of a viable IE etymology for Celtic **durno-*, which was the appropriate methodological response within the evidence available. The claim of the present paper is that a viable IE etymology is now available through the heteroclite framework. If that claim is correct, the substrate hypothesis is no longer necessary as an explanatory device.

The Demiraj reversal of the Albanian *dorë* lineage (1990, 1997) is engaged in §8 above; the citation chain on which it rests does not survive examination.

11. Scope and Limitations

Several clarifications are necessary to define the proposal's scope precisely.

The framework does not introduce a new PIE root. The bare **de-* and the locative **-r-* are independently attested morphemes; the proposal concerns their prior productive relationship. A reviewer committed to atomistic root reconstruction will resist the compositional analysis on principled grounds, and that resistance is coherent; the framework's claim is that the compositional reading is explanatorily superior to treating each reflex independently.

The reconstruction of **deh₃-* as the correct post-Anatolian verbal pre-form is not in dispute. The claim is about the internal morphological structure of that form — that **-h₃-* is a derivational extension suffix rather than an unanalyzable root consonant — and about the chronology of its addition. The phonological evidence for **h₃* in the post-Anatolian period is accepted; the question is only when and how the laryngeal was added.

The framework does not claim that all IE "hand" words descend from **de-r-*. PIE **ǵh₁esr-* (Greek χεῖρ, Hittite *keššar*) is a separate hand-word with its own well-attested comparative set, and the present analysis makes no claim about it. The Albanian *dorë* extension specifically argues that *dorë* does *not* belong to **ǵh₁esr-* (the standard view that Demiraj defends and that Belardi and Pokorny ruled out on phonological grounds), and instead belongs to the framework's **de-r-* family.

The Marstrand 1919 attribution recovers the lexical comparison and the ablaut analysis for the published lineage; it does not absolve the framework of the need to defend the formal mechanism (the heteroclite structure, the metathesis pathway, the Lubotsky revocalization). Marstrand's specific phonological argument — that Hittite *a* in *dār-na-áš* descends from IE **o* — is accepted in Hittite historical phonology. The framework's reconstruction is consistent with that finding.

The older derivation of δῶπον from **deh₃-ro-* remains formally available within standard handbooks. The Clayton/Blažek heteroclite framework is adopted here because it accounts for the Celtic cognates that the older derivation could not, and because it brings the cluster into alignment with Anatolian *tarawar*. Specialists may reasonably prefer the older analysis pending further evaluation of Clayton's reconstruction.

Latvian *dūre* is registered as supporting evidence but is not load-bearing; the Greek, Celtic, and Anatolian evidence is sufficient to ground the lexical claim independently.

12. The Remaining Papers

Paper 2 ("A Better Hand") reconstructs the PIE contrastive suffix **(t)ero-* compositionally as **de-* + **-r-* + **-o-*, and develops the numeral **d̥uo-/d̥ui-* "two" as a parallel derivational outcome on the same root. The bilateral character of **(t)ero-* — its consistent function of marking "one of a pair" across the family — is argued to be a morphological inheritance from the **-u-* element of pairedness, not a typological consequence of bilateral body symmetry. Dunkel's (2014) published etymology unifying the prefixal **ui-* and the numeral **d̥ui-*, together with his reconstruction of the inherited compound **ui-tero-* (Vedic *vítarām*, OCS *vŭtoru* "the second"), provides the external support.

Paper 3 ("The Bare Hand") establishes **de-* as an independently attested PIE directional particle and decomposes the give-verb **deh₃-* as **de-* + *verbal extension* **-h₃-*. The strongest evidence for this bimorphemic decomposition is a class-by-class asymmetry in the Greek root-aorist 2pl imperative paradigm: roots the framework analyzes as bimorphemic (including **deh₃-* → δότε) preserve a bare-CV form in the 2pl, while roots the framework does not analyze as bimorphemic (including **steh₂-* → στήτε) preserve the laryngeal-conditioned long vowel throughout. The asymmetry is descriptively established in the standard grammatical tradition; the framework provides a structural account of why it should exist.

Paper 4 ("The Hand That Takes") derives Hittite *dā-* "to take" from a pre-Anatolian directive univerbation **de + a*. The proposal fits a documented productive class in attested Hittite: the Set 1 directional system in which a local element is followed by the directional particle **-a* and contracts to a fixed adverb or preverb (*anda* < **en + a*, *katta* < **k̥mt + a*, *parā* < **pro + a*). The proposal resolves the take/give polarity — the fact that Anatolian's reflex of **deh₃-* means "take" while post-Anatolian reflexes mean "give" — as a structural consequence of the directional **a* "toward the speaker" combining with the hand-element to yield "extend toward oneself" = "take." The Hittite/Luwian *tarāwi(ya)-* "to hand over, deliver" and the corresponding noun *tarawar* (§9

above) provide independent Anatolian witness to the framework's heteroclitic gift-noun branch at Norbruis's reconstructed intermediate stage.

13. Probability Assessment

We assess the probability of the Paper 1 thesis — that the comparative cluster reflects two derivational siblings on the same pre-PIE hand-element **de-* — at approximately 40–50%, conditional on its own evidence considered independently of Papers 2–4. This is a revision upward from the 30–40% estimate of v33, reflecting the recovery of the Marstrander 1919 attribution and the integration of the Anatolian **tarawar*/tarāwi(ya)-* witness.

In favour of the analysis: the lexical comparison itself rests on a continuously-published lineage from Marstrander 1919 through Pokorny 1959 to Blažek 1999 and the modern Greek etymological dictionaries (Frisk, Beekes 2010, DGE), with three independent twentieth-century departures (Schrijver, Beekes on Celtic, Matasović) each addressable on its own terms. The framework's formal mechanism — the heteroclitic ***wṛ-/w(e/o)n-* structure with regular **wṛ > *ru* metathesis — has independent support in two published treatments (Clayton 2023, Blažek 1999) that arrive at the heteroclitic class from different empirical bases. The OKhotanese *durna-* parallel (Bailey 1961) is the closest phonological analog for the Celtic development in the literature. The unification of *δωρ-* and *δαρ-* as strong and weak stems of a single heteroclitic paradigm provides a structural account where Schrijver's label "obscure" has stood for thirty years. The Anatolian witness **tarawar*/tarāwi(ya)-* (Blažek 1999, Tischler 1991, Melchert 1993) provides published Anatolian evidence at Norbruis's reconstructed semantic intermediate stage. The Albanian *dorë* extension rests on a 127-year published lineage that the standard rejection (Demiraj 1990, 1997) does not adequately defend.

Against: the compositional body-part noun **de-r-* is non-standard as a synchronic PIE reconstruction, even though its components (**de-* and **-r-*) are independently attested. The Celtic derivation involves more steps than earlier proposals, even if each step is independently motivated. The broader framework across all four papers requires a productive pre-PIE morphological system whose existence can only be evaluated by assessing the evidence together.

The probability at the level of the whole project — integrating the conditional probabilities across Papers 1–4 — is in the range 2–5%, with much of the evidence supporting multiple strata simultaneously rather than each stratum independently. The marginal figure is somewhat higher than the v33 estimate (1–4%) because the recovery of Marstrander's attribution and the Anatolian witness raises the conditional probability of Paper 1 specifically. We continue to regard the central claim as more likely wrong than right at the

marginal level, and the evidential structure as sufficiently dense and the proposal sufficiently testable to warrant publication for scholarly examination.

Authorship and Contributions

Chavis's contributions. The central hypothesis originates with Chavis: the identification of the dual attestation of Greek δῶρον — gift and palm-breadth — as evidence of a single etymological family was the project's founding insight. Chavis directed the research agenda at every stage: the hypothesis that **(t)ero-* descends from the same root; the identification of *ambidexter* as the morphologically transparent compound witness for Paper 2; the question "whence comes the *r* in **-tero-*?" that opened the compositional analysis of Paper 3; the hypothesis that Hittite *dā-* preserves a pre-Anatolian construction; and the decision to incorporate Clayton (2023) as the formal linchpin of the lexical argument. The decision to pursue the Albanian *dorë* extension despite Demiraj's published rejection, and to ground the framework's analysis in direct examination of the citation chain Demiraj invokes, is Chavis's. The decision to integrate the Marstrander 1919 attribution and the Anatolian **tarawar*/tarāwi(ya)-* witness into Paper 1's central architecture — restructuring the paper around the published lineage rather than around a defensive engagement with the standard view — is Chavis's. All fundamental decisions about scope, framing, and direction across all four papers rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude's contributions. Claude's contributions are primarily in source synthesis, exposition, and analytical development. Across the project, Claude located and read the relevant published literature: Marstrander 1919 *Caractère indo-européen de la langue hittite*, p. 128 + footnote 1, with the Hesychian apparatus; Blažek 1999 in *Tocharian and Indo-European Studies* 8, with the heteroclitic **dwor-eH₂-wr₂-w(e)n-* reconstruction and the Anatolian *tarawar* witness; Tischler's *Hethitisches Etymologisches Glossar*, including the Marstrander 1919 attribution preserved at HEG III/9: 195 (Lieferung 9, 1983) for the *tarna-* measure-noun; Clayton (2023) on the heteroclitic class; Lipp (2009), Bauhaus (2019), and Kahl (2024) on the locative **-r-*; Norbruis (2021) on the semantic chronology of **deh₃-*; Stifter (2022) on the Bronze of Novallas; Lubotsky (1994, 1997) on the phonological mechanisms; Dunkel (2014) on the **ui-tero-* compound; Belardi 1948 on the Albanian **-sr-* impossibility; La Piana 1937 and 1942 (the latter via Frisk, Beekes, and DGE) on the **dōrom* derivation; the DGE laryngealized reconstruction **deh₃r-* that essentially identifies the framework's **de-h₃-r-*; Tagliavini *L'albanese di Dalmazia* directly, establishing that AD 101 does not contain the rejection Demiraj cites; Kloekhorst (2008,

2013) and Hoffner & Melchert (2008) on Hittite; Lejeune (1961) and van Hes (2021) on Mycenaean -δε; and ancillary sources throughout. Claude drafted the prose of all four papers from Chavis's structural and substantive direction; identified Bauhaus's locative-directive paradigmatic pairs as published structural precedent for Papers 3–4; integrated the architectural alignment across all four papers at each revision stage; and proposed the present restructure of Paper 1 around the published lineage (Marstrander → Pokorny → Blažek/Clayton) once the Marstrander attribution surfaced. The probability estimates in all four papers are Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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